SYNOPSIS

Concord: An Experimental Study of Autonomous Negotiation Strategies Between Large Language Model Agents

A Mini Project

Submitted in partial fulfillment of the requirement for the award of Degree of Bachelor of Technology in Computer Science & Engineering

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ABSTRACT

Negotiation is a fundamental process underlying commerce and decision-making, and the emergence of capable open-weight Large Language Models (LLMs) makes it possible to study negotiation as a computational, reproducible phenomenon. This project, Concord, places two independently prompted instances of Google's Gemma 4 model on opposite sides of a bargaining scenario — a buyer and a seller, each holding a private reservation price — and allows them to negotiate autonomously over several conversational turns.

The system varies agent persona (cooperative, aggressive, neutral) and information symmetry (shared vs. private reservation values) across a set of controlled experimental conditions, logging every negotiation as structured data. The resulting transcripts are analyzed using metrics such as deal success rate, rounds to convergence, surplus split, and Pareto efficiency to determine how prompting strategy and information asymmetry shape negotiation outcomes. The project aims to produce a working negotiation simulation framework, a dataset of experimental transcripts, and a comparative analysis of LLM negotiation behaviour under different strategic conditions.

Keywords: Large Language Models, Multi-Agent Systems, Automated Negotiation, Gemma 4, Prompt Engineering, Game Theory

CHAPTER 1: INTRODUCTION

1.1 Overview

Negotiation is a goal-directed conversational process in which two or more parties with partially conflicting interests attempt to reach a mutually acceptable agreement. Historically, automated negotiation has been studied through game-theoretic bargaining models and rule-based software agents. The recent emergence of instruction-tuned Large Language Models capable of sustained, contextual, multi-turn dialogue opens a new avenue: negotiation can now be carried out entirely in natural language by autonomous AI agents, without hand-crafted negotiation protocols.

This project, Concord, uses Google's open-weight Gemma 4 model to build and study such a system. Two independent Gemma 4 agents are configured with opposing roles (buyer and seller), private information (reservation prices), and a persona (cooperative, aggressive, or neutral), and are then allowed to negotiate over a single item through alternating natural-language offers and counter-offers. The project is not intended to build a single production chatbot, but rather a small, controlled experimental framework: by systematically varying persona and information conditions across many simulated negotiations, the project studies how these factors influence whether a deal is reached, how quickly, and how fairly the resulting surplus is divided between the two sides.

Gemma 4 was selected as the underlying model because it is open-weight and can be run without recurring per-token API costs, supports native system-role prompting for structured agent configuration, and includes configurable reasoning modes suited to multi-step strategic dialogue — making it a practical and reproducible choice for a semester-length academic project.

CHAPTER 2: PROBLEM IDENTIFICATION & SCOPE

2.1 Problem Domain

Most publicly available LLM-based systems are designed for single-agent, cooperative, human-facing conversation. There is comparatively little accessible, reproducible experimental work examining how two independently prompted instances of the same open-weight LLM behave when placed in an adversarial, goal-driven, multi-turn negotiation against each other. As autonomous AI agents increasingly transact and interact with other AI agents (in automated procurement, price bargaining assistants, and multi-agent systems generally), it becomes important to understand systematically how such agents behave strategically: whether they concede too quickly, exploit information asymmetry, negotiate to an impasse, or reach economically inefficient agreements.

The problem this project addresses can be stated as: How do prompting strategy (persona), and information asymmetry (private vs. shared reservation values) affect the outcome, efficiency, and fairness of a negotiation carried out between two Gemma 4 LLM agents?

2.2 Solution Domain

The proposed solution is a controlled negotiation-simulation framework built around a single, well-bounded scenario — buyer-seller price negotiation over one item — so that outcomes remain comparable and interpretable across experimental runs. Two Gemma 4 agent instances are configured with distinct system prompts encoding role, private reservation price, and persona, and an orchestrator script manages alternating turns until a deal is reached, a walk-away condition triggers, or a maximum round limit is hit.

The scope of the project covers the following experimental conditions: persona variation (cooperative vs. aggressive vs. neutral) and information-symmetry variation (both reservation prices hidden vs. one agent aware of the other's constraint), run across approximately 20–30 simulated negotiations per condition. All transcripts are logged as structured JSON and analyzed using Python to compute deal success rate, rounds to convergence, surplus split, and Pareto efficiency. The scope deliberately excludes multi-item or multi-party negotiations, real-time human-in-the-loop negotiation, and fine-tuning of the base model — these are identified as possible future extensions rather than part of the current project.

CHAPTER 3: SOFTWARE ENGINEERING APPROACH

3.1 Software Model Used

3.1.1 Description

The project follows an Iterative and Incremental development model. The system is built up in small, working increments — starting with a minimal single-scenario, single-condition negotiation pipeline between two Gemma 4 agents, and then incrementally adding persona variation, information-asymmetry variation, structured logging, metric computation, and finally a visualization/reporting layer. Each increment produces a runnable version of the system that is tested before the next feature is added.

3.1.2 Reason for Use

An iterative and incremental model is well suited to this project because the requirements — particularly the exact set of experimental conditions and metrics worth measuring — are expected to be refined as early negotiation runs are observed. Building the core two-agent pipeline first and validating it before layering on experimental conditions reduces the risk of late-stage rework, keeps the system demonstrable at every stage of the semester, and fits naturally within the fixed, incremental timeline of a semester-long academic minor project.

3.2 Platform Specification

3.2.1 Hardware Specification

- Processor: Intel Core i5 (or equivalent) or higher
- RAM: 8 GB minimum (16 GB recommended for smoother local inference/testing)
- Storage: 10 GB free disk space for project files, logs, and datasets
- Internet connection for accessing the hosted Gemma 4 inference endpoint (if not run fully locally)

3.2.2 Software Specifications

- Operating System: Windows 10/11 or Linux (Ubuntu)
- Language Model: Gemma 4 (open-weight, instruction-tuned variant)
- Programming Language: Python 3.x
- Libraries: pandas (metric computation and analysis), Streamlit (optional visualization dashboard)
- Data Format: JSON for structured negotiation transcript storage
- Development Tools: VS Code / Jupyter Notebook, Git for version control

CHAPTER 4: EXPECTED OUTCOME

- A working negotiation simulation framework in which two Gemma 4 agents can be configured with different personas and information conditions and made to negotiate autonomously.
- A dataset of simulated negotiation transcripts covering all experimental conditions (persona × information symmetry).
- A comparative analysis showing how persona and information asymmetry affect deal success rate, rounds to convergence, surplus split, and Pareto efficiency.
- A brief demonstration interface capable of displaying a negotiation unfolding turn by turn, along with summary charts across conditions.
- A final project report documenting the methodology, experimental results, and conclusions drawn about LLM negotiation behaviour.

REFERENCES

[1] Google DeepMind, “Gemma 4: Byte for byte, the most capable open models,” Google AI Blog, 2026.

[2] A. Rubinstein, “Perfect Equilibrium in a Bargaining Model,” Econometrica, vol. 50, no. 1, pp. 97–109, 1982.

[3] Relevant literature on LLM-based automated negotiation and multi-agent dialogue systems (to be expanded with specific papers in the final report, in consultation with the project supervisor).